



how to evaluate by MonteCarlo methods the bias and variance of an unf



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AI Overview

To evaluate the bias and variance of an unfolding method with signal and background contributions using Monte Carlo (MC) methods, generate multiple pseudo-datasets by varying both the total event counts per bin (via Poisson sampling around the expected signal plus background) and independently fluctuating the background estimate, then apply the unfolding algorithm to each replica (Cowan, 1998).

Monte Carlo Evaluation Procedure

- Define a reference "true" neutrino flux as a function of neutrino energy, say $f_{true}(E_\nu)$, and map it via the coherent elastic neutrino-nucleus scattering kernel $K(E_r, E_\nu)$ to predicted recoil bin rates $\mu_s(E_r)$. Add a background rate vector μ_b .
- Compute expected total bin counts as $\lambda_i = \mu_{s,i} + \mu_{b,i}$ for each recoil energy bin i .
- Generate M mock datasets. For each dataset m , draw total binned observed counts $n_i^{(m)}$ from $\text{Poisson}(\lambda_i)$.
- Independently generate a perturbed background estimate $b_i^{(m)}$ for each dataset, either by drawing from $\text{Poisson}(\mu_{b,i})$ or from a Gaussian distribution representing the background uncertainty covariance matrix.
- Subtract $b_i^{(m)}$ from $n_i^{(m)}$ (or supply both to a background-aware unfolding routine) to yield the net signal estimate in data space.
- Unfold each mock dataset m to obtain an estimate of the flux in energy bins, denoted as $f^{(m)}(E_\nu)$ or vector $x^{(m)}$.
- Estimate the expectation value via the MC sample mean $\bar{x} = (1/M) \sum_{m=1}^M x^{(m)}$.
- Compute the empirical bias vector as $\text{bias} = \bar{x} - x_{true}$, and compute the empirical covariance matrix as $S = (1/(M-1)) \sum_{m=1}^M (x^{(m)} - \bar{x})(x^{(m)} - \bar{x})^T$.

Minimized Quantity and the Bias-Variance Tradeoff

The combined quantity that regularized unfolding methods (such as Tikhonov regularization) minimize is the Mean Squared Error (MSE), or more precisely, the expected total risk $E[\|\hat{x} - x_{true}\|^2]$, which decomposes directly into the squared bias term $\|\text{bias}\|^2$ plus the sum of the variances (the trace of the covariance matrix) $\text{tr}(S)$ (Hansen, 1998). By introducing a small, controlled amount of smoothing (increasing the bias slightly via the regularization parameter α), the inverse problem is stabilized, leading to a dramatic, often exponential drop in the variance $\text{tr}(S)$ and a net reduction in total error (Blobel, 1985).

How Tikhonov Regularization Reduces Variance

Tikhonov regularization modifies the least-squares chi-squared objective by adding a penalty term $\alpha^2 \|Lx\|^2$, where L is a discrete derivative operator. Expressed via the Singular Value Decomposition (SVD) of the response matrix, each singular value σ_j is modified by a filter factor $f_j = \sigma_j^2 / (\sigma_j^2 + \alpha^2)$. When $\sigma_j \ll \alpha$, the unregularized inverse contributions $1/\sigma_j$ that blow up and amplify statistical noise are heavily suppressed by the α^2 denominator. This dampening removes high-frequency components, causing an

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Other methods are possible, for example $f(t)$ can be discretized by a superposition of B-splines [3] which...

arXiv

Empirical Bayes estimation and bias-corrected uncertainty ...

Unfolding typically proceeds in two steps: one first produces a regularized point estimate of the unknown...

exponential or rapid reduction in the variance of the solution coefficients at the cost of a modest, smooth increase in the squared bias.

Behavior with Respect to α

- As α increases, the filter factors f_j shrink further for small singular values, causing the variance $\text{tr}(S)$ to decrease monotonically toward zero.
- Simultaneously, the penalty forces the solution to become smoother or closer to a default model, which increases the systematic deviation from the true distribution, meaning the squared bias increases monotonically from zero.

Variance-Covariance Matrix vs. Trace

The term "variance-covariance matrix" refers to the full matrix S where the diagonal

elements represent the variance of each individual energy bin and the off-diagonal elements represent the covariances between different energy bins. The variance of a single bin is a scalar number, but when characterizing the uncertainty of an entire unfolded multi-bin spectrum simultaneously, a matrix is required. The trace of the covariance matrix, $\text{tr}(S) = \sum_j S_{jj}$, sums up the individual variances of all bins, thereby incorporating the diagonal components of the uncertainty, but it discards the off-diagonal correlation structures. Using the full matrix retains information on how an error in one energy bin correlates with fluctuations in neighboring energy bins.

Evaluation Error as a Function of M

The MC statistical error in estimating the mean or variance from M independent mock datasets scales inversely with the square root of the number of trials. For an estimated mean $\bar{x}$, the standard error of the mean for each component is given by $\sigma_{\bar{x}} = s/\sqrt{M}$, where s is the sample standard deviation of the unfolded estimates across the M runs. Similarly, the standard error for the variance estimator itself is approximately $\sigma_{s^2} \approx s^2 \sqrt{2/(M-1)}$.

Verified References

- Blobel, V. (1985). Unfolding methods in high-energy physics. *CERN Report*, 85-09, 88-126.
- Cowan, G. (1998). *Statistical Data Analysis*. Oxford University Press. (See chapters on Monte Carlo techniques and propagation of errors).
- Hansen, P. C. (1998). *Rank-Deficient and Discrete Ill-Posed Problems: Numerical Aspects of Linear Inversion*. SIAM. (Provides the mathematical treatment of Tikhonov filter factors and the SVD bias-variance decomposition).

Further Exploration

- Review foundational unfolding algorithms and regularization strategies in the CERN Document Server.
- Explore advanced Bayesian and frequentist comparisons in the arXiv repository.  Home | CERN +1





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<https://indico.cern.ch/attachments/blobel>

Unfolding Methods in Particle Physics* - Indico

by V Blobel · Cited by 84 — The transformation to the underlying true distribution is called **unfolding** in particle physics and belongs to the class of linear inverse problems. If the ...
12 pages



GitHub

<https://goodboychan.github.io/2020/06/16/03-Monte...>

Monte Carlo Policy Evaluation

Jun 16, 2020 — Calculate MC empirical average return from episodes which visit the state s to approximate **value function**: $V_{\pi}(s) = E_{\pi}[G_t | s_t = s]$... [Read more](#)



Medium · victor_

5 years ago

The basics of Monte Carlo integration - victor - Medium

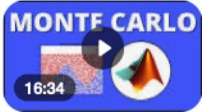
Monte Carlo integration is a **technique** for numerical **integration** using random numbers. Basic concept of the **Monte Carlo** estimator. Let's try to ... [Read more](#)
Missing: bias unfolding signa convolution; kernel background? (verified-



Reddit · r/matlab

3 comments · 4 years ago

Monte-Carlo Integration is a personal favorite - here's a step-by ...



Monte Carlo Integration is a numerical approach to integrals. We ignore the analytical solution and we're going to use random points to estimate an integral.

Missing: bias variance unfolding signa convolution; kernel background? (verified- references procedure



Uni Ulm

https://www.uni-ulm.de/lehre/Lecture_Notes

Methods of Monte Carlo Simulation

by T Brereton · Cited by 6 — **Monte Carlo methods** are tools for solving problems using random numbers. in order to make **Monte Carlo methods** more **accurate** we should aim to find a **Monte...**
61 pages



Wikipedia

https://en.wikipedia.org/wiki/Monte_Carlo_integrati...

Monte Carlo integration

Monte Carlo integration is a **technique** for numerical **integration** using random numbers. **Monte Carlo** randomly chooses points at which the integrand is **evaluated**. ...
Missing: bias unfolding rate convolution; kernel unfold) background? (verified-



Physically Based Rendering: From Theory to Implementation

https://pbr-book.org/Monte_Carlo_Integration/The_...

13.2 The Monte Carlo Estimator

13.2 The **Monte Carlo** Estimator. We can now define the basic **Monte Carlo** estimator, which approximates the **value** of an arbitrary **integral**. [Read more](#)

Missing: bias unfolding signa convolution; kernel function unfold) (verified- invented) references procedure you



Mathematics Stack Exchange

<https://math.stackexchange.com/questions/variance-...>

Variance of Estimator (Monte Carlo Integration)

Jun 18, 2018 — Copying this question directly from another stack-exchange site, since didn't receive

any good answers there. So I was reading this paper by Lafortune, ...

1 answer · Top answer: This computation really has nothing to do with the Monte Carlo integration co...



PubMed Central (PMC) (.gov)

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9005059>

Monte Carlo Simulation Approaches for Quantitative Bias ...

by HR Banack · 2021 · Cited by 84 — Quantitative **bias** analysis **can** be used to empirically **assess** how far study estimates are from the truth (i.e., an **estimate** that is free of **bias**).

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